Optimal Bitcoin Trading Strategy - Final

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Introduction

The cryptocurrency market, particularly Bitcoin, presents unique challenges for forecasting due to its high volatility and sensitivity to various external factors. This project explores the application of Machine Learning and Stochastic Processes to formulate a trading strategy for bitcoin. Combining a computer science approach with economic forecasting principles, I aim to develop a framework that can effectively predict optimal trading actions (buy, sell, or hold) and their respective magnitudes (for buy and sell) based on historical market data and related financial indicators.

Research Question

How can we develop a robust daily decision-making framework for Bitcoin trading that effectively determines optimal actions (buy, sell, or hold) and their magnitude to maximize returns while managing risk across different market conditions?

Starting with an initial budget B_0, an investor makes a sequence of daily trading decisions over a period of T days, denoted as $D = \{d_1, d_2, ..., d_T\}$. For each day t, the decision $d_t \in \mathbb{R}$ represents the amount of Bitcoin to trade, where:

- $d_t > 0$ indicates buying d_t units of Bitcoin
- $d_t < 0$ indicates selling $|d_t|$ units of Bitcoin
- $d_t = 0$ indicates holding the current position

Given the daily Bitcoin prices $P = \{p_1, p_2, ..., p_T\}$, the investor's portfolio value V_t after day t can be expressed as:

$$V_t = B_t + Q_t \cdot p_t$$

Where B_t is the remaining cash budget and Q_t is the quantity of Bitcoin held after day t. These values are updated according to each decision:

$$B_t = B_{t-1} - d_t \cdot p_t - c(d_t, p_t)$$
$$Q_t = Q_{t-1} + d_t$$

With $c(d_t, p_t)$ representing transaction costs associated with the trade.

The objective is to find the optimal decision sequence D^* that maximizes the final Sharpe Ratio while maintaining acceptable drawdown levels. Here, each decision where each decision implicitly relies on forecasting future Bitcoin price movements based on available market information up to day $t-1$. Note, at any given time there may be excess cash that the algorithm has decided not to invest into bitcoin. Return on this excess cash will be based on that day's risk free rate, which would provide a lower bound on average return from alternate investments that are independent of bitcoin.

Data

Data Description

The analysis will utilize a dataset of daily financial indicators from 2015 to present, sourced through the cctx and yfinance python libraries:

- **Bitcoin OHLCV Data**: Sourced directly from Coinbase cryptocurrency exchange via the CCXT library, providing daily Open, High, Low, Close prices and Volume information
- **Gold Prices**: Represented by the SPDR Gold Shares ETF (GLD), obtained from Yahoo Finance, including both price and volume data
- **Oil Prices**: Tracked using the United States Oil Fund (USO), also obtained from Yahoo Finance, with price and volume metrics
- **Risk-Free Rate**: 3-month Treasury yield (^IRX) sourced from Yahoo Finance as a proxy for the risk-free interest rate
- **Equity Market**: S&P 500 Index (^GSPC) data from Yahoo Finance, including both price and trading volume to represent broader market movements and liquidity
- **Cryptocurrency Ecosystem**: Bitwise 10 Crypto Index Fund (BITW) data from Yahoo Finance, with both price and volume data to provide context within the wider cryptocurrency market
- **Transaction costs:** Transaction costs are dynamic and based on multiple factors like volume, spread, market conditions, etc. I will try to approximate this using binance's fee schedule which is based on 30 day trading volume.

Collection

All data collection is automated by a python script, which handles API connections, synchronizes data from different sources, and implements appropriate cleaning procedures. The script performs several key functions:

- Fetches Bitcoin data via CCXT with proper rate limiting

- Collects complementary financial data through Yahoo Finance
- Aligns data points to ensure temporal consistency
- Handles missing values through forward-fill (mainly to fill in weekend prices for all variables not related to bitcoin)
- Merges datasets into a unified dataframe and saves the resulting csv file.

Dependent and Independent Variables

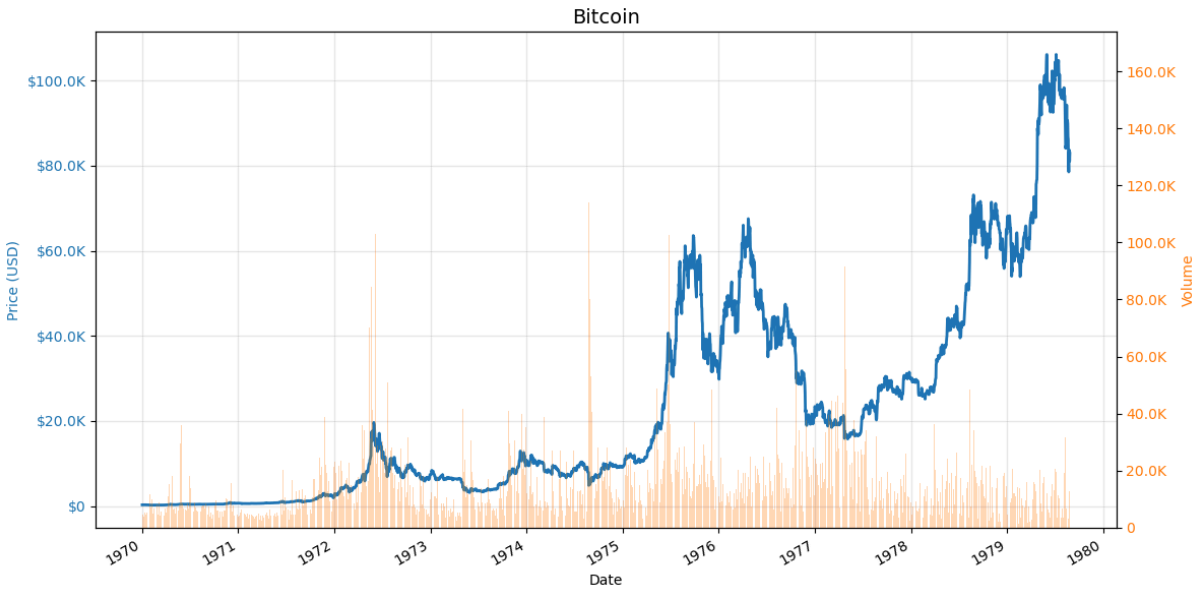
The primary dependent variable will be Bitcoin's daily close price, while independent variables will include:

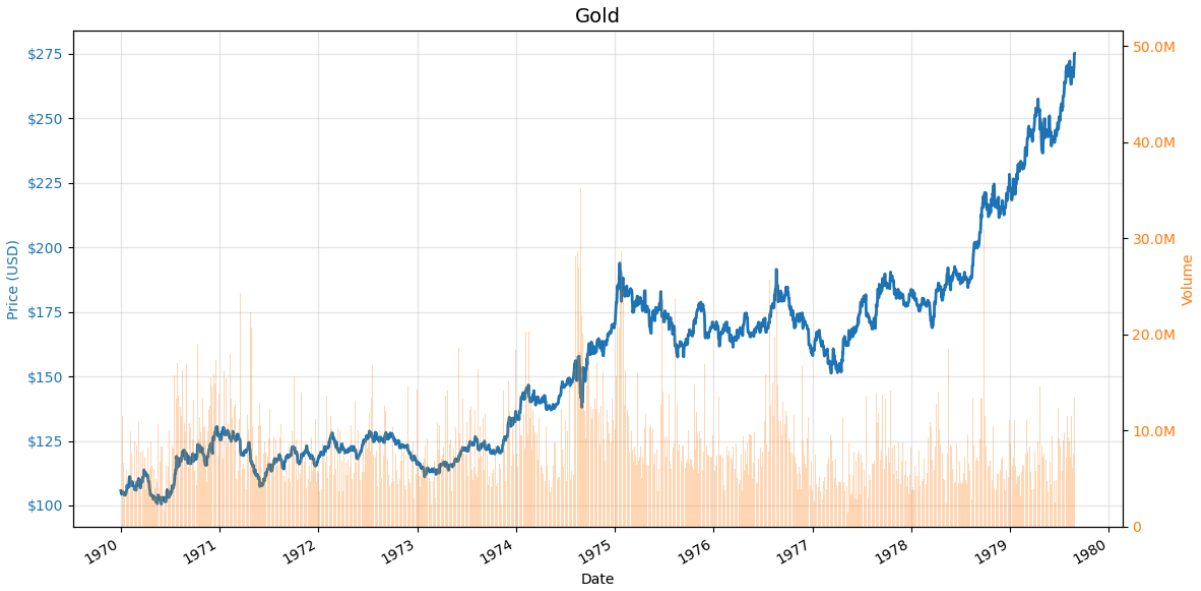
- Lagged values of Bitcoin prices
- Volume metrics for Bitcoin and related assets (important for liquidity analysis)
- Technical indicators derived from price and volume data (moving averages, relative strength index, Bollinger bands, volume oscillators)
- Price and volume correlations between Bitcoin and traditional financial assets
- Volatility measures derived from high-low ranges

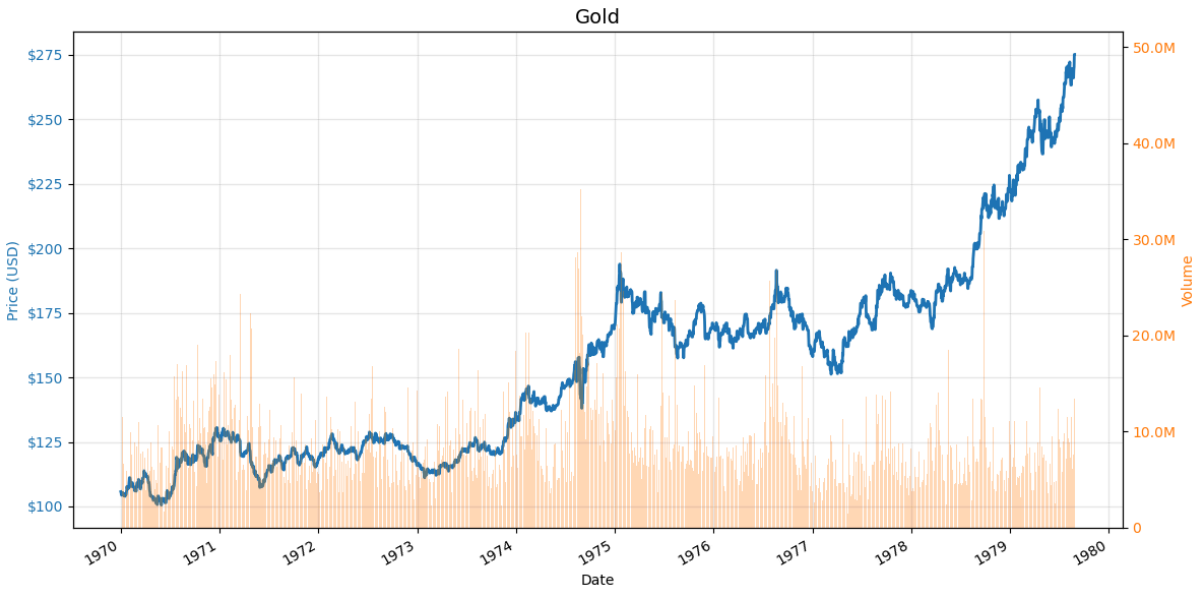
The dataset will be split chronologically into training (70%), validation (15%), and testing (15%) sets to properly evaluate the forecasting models without look-ahead bias. All price data will be normalized to facilitate model training.

Key Metrics

Visualizations




Gold
Price (USD)
$275
$250
$225
$200
$175
$150
$125
$100
Volume
50.0M
40.0M
30.0M
20.0M
10.0M
0
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
Date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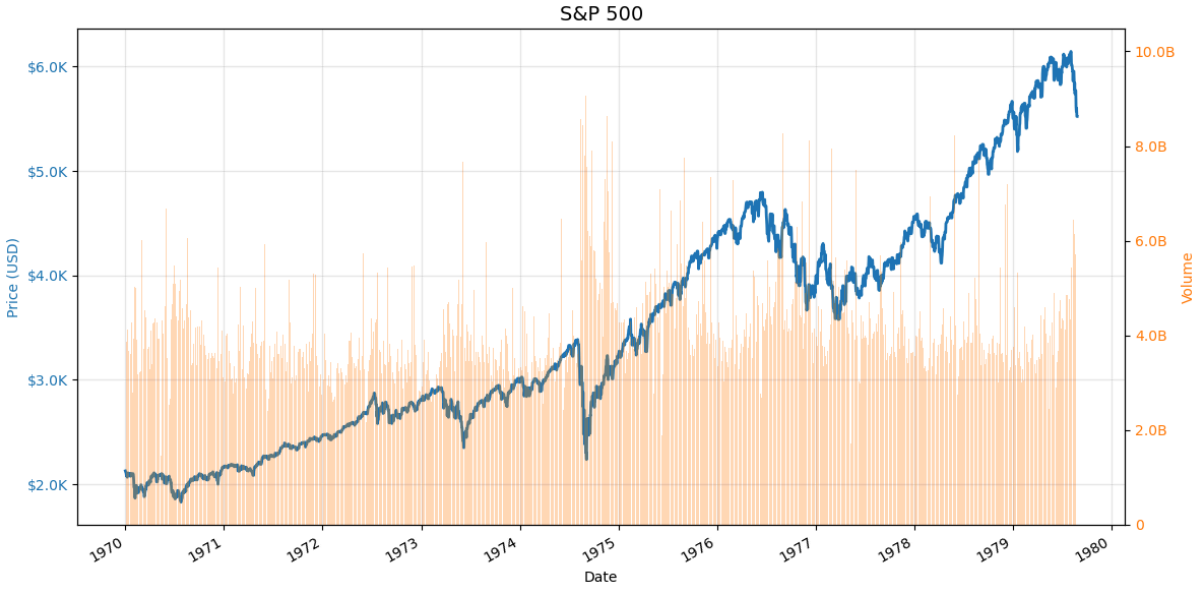
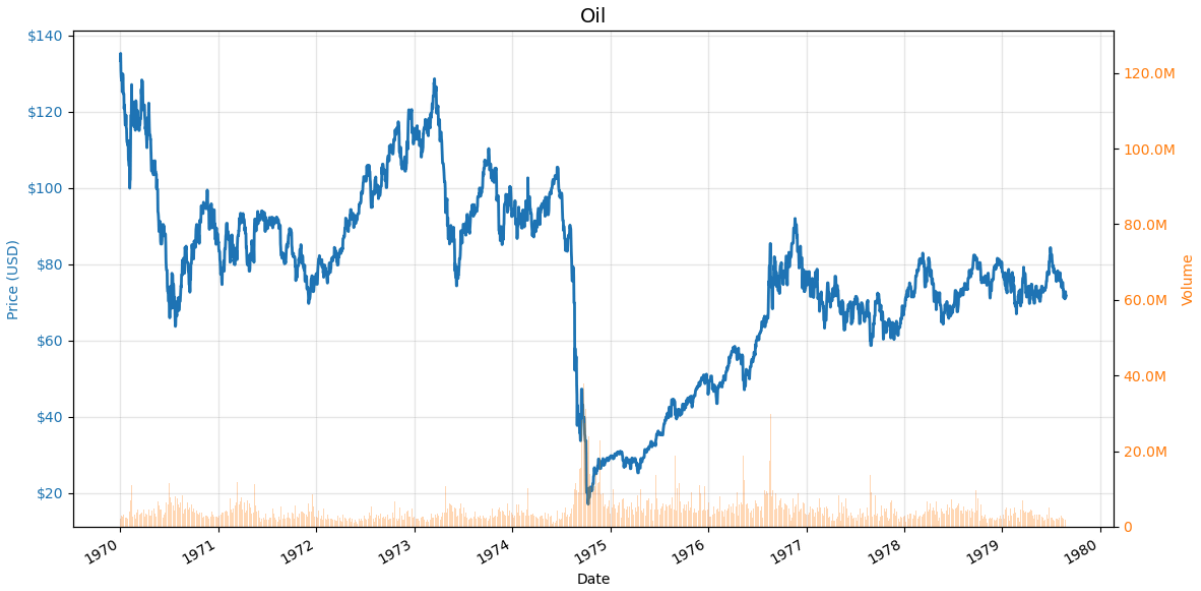

S&P 500
Price (USD)
$6.0K
$5.0K
$4.0K
$3.0K
$2.0K
Volume
10.0B
8.0B
6.0B
4.0B
2.0B
0
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
Date



Summary Statistics

	btc_close	btc_volume	gld_price	gld_volume	uso_price	uso_volume	sp500_price	sp500_volume
mean	22819.45	14398.3	154.89	8646902.16	76.76	4424337.86	3482.33	4068799926.26
std	24479.6	11759.82	38.01	4583017.3	23.17	4418623.75	1118.29	1076005442.73
min	211.16	683.8	100.5	1436500.0	17.04	431738.0	1829.08	1296530000.0
25%	4004.59	6739.0	121.37	5718725.0	66.96	2505575.0	2581.02	3434400000.0
50%	10789.38	11355.56	156.28	7550900.0	77.94	3551969.0	3222.33	3841910000.0
75%	37266.84	18243.23	177.68	10378075.0	91.68	4999425.0	4327.62	4431900000.0
max	106159.26	165542.81	275.13	49139000.0	135.28	124913013.0	6144.15	9976520000.0

Timeline

Data Preprocessing

- Exploratory Data Analysis (EDA)
- Time series decomposition to understand trends, seasonality, and noise
- Feature engineering to create technical indicators
- Correlation analysis to identify relationships between Bitcoin and other assets

Model Development:

Reinforcement Learning Approach:

- Implementing a Deep Q-Network (DQN) where the state will be represented by recent price movements and technical indicators
- The action space will include discrete options for buying, selling, or holding with different magnitude levels.
- Reward function will be based on risk-adjusted returns
- Potentially also exploring policy gradient approaches

Stochastic Process Approach:

- Modeling Bitcoin price movements using geometric Brownian motion
- Implementing Monte Carlo simulations to generate potential price paths
- Using the simulation results to estimate probability distributions of future prices
- d_t's will depend on the resulting distribution.

Evaluation

The project will evaluate the effectiveness of our trading strategy's decisions using the following performance metrics that directly assess how well our algorithm determines the optimal quantities (d_i values) to buy, sell, or hold:

Performance Metrics

Sharpe Ratio

Measures the excess return per unit of risk generated by our trading decisions. It is given by the formula:

$$sharpe = \frac{R_p - R_f}{\sigma_p}$$

where:

- $R_p$is the annualized portfolio return resulting from our sequence of trading decisions (d_i)
- R_f is the risk-free rate (3-month Treasury yield)
- σ_p is the annualized standard deviation of our portfolio's periodic returns

Calculation process:

1. For each time period t, the algorithm generates a decision d_t (quantity to buy/sell)
2. This updates our Bitcoin holdings Q_t and cash balance B_t
3. Portfolio value at each step: $V_t = B_t + Q_t \times p_t$
4. Calculate periodic returns: $r_t = \frac{V_t - V_{t-1}}{V_{t-1}}$
5. $R_p = \left(1 + \frac{\sum_{t=1}^{n} r_t}{n}\right)^{365} - 1$
6. $\sigma_p = \sqrt{\frac{\sum_{t=1}^{n}(r_t - \bar{r})^2}{n-1}} \cdot \sqrt{365}$

Maximum Drawdown

Measures the largest peak-to-trough decline in portfolio value resulting from our trading decisions. It is given by the formula:

$$MaxDD = \frac{\max\left(\max(V_s) - V_t\right)}{\max(V_s)} \qquad \forall s \leq t$$

Where V_t is the portfolio value at time t, directly resulting from our sequence of decisions.

Calculation process:

1. For each time period t, our algorithm generates a decision d_t (quantity to buy/sell)
2. This updates our Bitcoin holdings (Q_t) and cash balance (B_t)
3. Calculate portfolio value at each step: $V_t = B_t + Q_t \cdot p_t$
4. Track the running maximum portfolio value: $Peak_t = \max(V_1, V_2, ..., V_t)$
5. Calculate drawdown at each step: $DD_t = (Peak_t - V_t)/Peak_t$
6. Maximum drawdown is the largest drawdown observed: $MaxDD = \max(DD_1, DD_2, ..., DD_T)$

A lower maximum drawdown indicates our algorithm is generating d_t values that effectively reduce exposure before significant price drops

This directly evaluates if our model correctly identifies high-risk periods and adjusts position sizes accordingly

Benchmark Comparison

Outperformance vs. Buy-and-Hold

Compares the performance metrics (sharpe ratio and maximum drawdown) of our dynamic trading strategy against a simple buy-and-hold approach

- Buy-and-hold would be represented by $D_h = \{\frac{B_0}{p_1}, 0, 0, ..., 0\}$
- Our strategy generates a different sequence of d_t values that varies exposure over time
- The resulting Sharpe ratio and Drawdowns will be different and this comparison will be the final evaluation

Trade Analysis

Examine individual trading decisions to evaluate their effectiveness through the following metrics:

- Win rate: Percentage of d_t decisions that contributed positively to portfolio value
- Profit factor: Sum of gains from profitable decisions divided by sum of losses
- Average profit/loss per trade: Mean portfolio value change resulting from each non-zero d_i
- Average holding period: Typical duration between buy and sell decisions

This analysis helps identify whether our algorithm makes consistently good quantity decisions or relies on a few outsized winners.

These metrics will be mainly used on the test set to understand what the model is really doing so I might find ways of improving it or making necessary changes.

Tentative Targets

Prior to model development I am setting the following targets (which are subject to adjustment) for evaluating whether the approach was successful or not:

1. Sharpe ratio at least 30% higher than a buy-and-hold strategy over the test period.
2. Maintains a maximum drawdown at least 20% lower than a buy-and-hold strategy.
3. Win rate of at least 55%
4. Profit factor of at least 1.5 will indicate success.